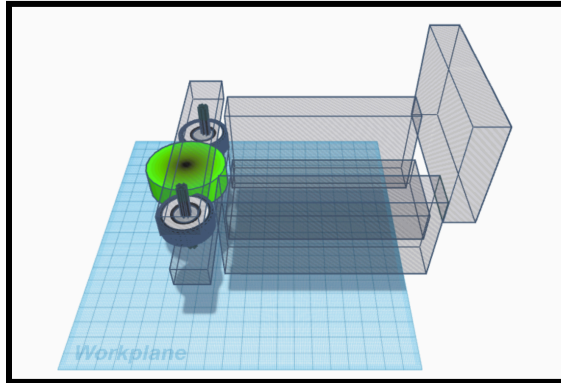


Element D: Concepts/Designs to Contribute to Solution

Brainstorming Sketches



Feedback:

For the flywheel disc launcher, the user specifically suggested for the mechanism to be smaller so students can easily use it and portable for teachers. Another concern that the user voiced was the length of the battery life where they would not need to be constantly replacing batteries.

Design Matrix:

	Frisbee Catapult	Single Flywheel Launcher	Double Flywheel Launcher
Mobility- Easy to move/portable	5	3	2
Repairability - Can the solution be easily fixed if a problem were to occur (mechanical or electrical)	5	3	3
Performance - capability launch frisbee a certain distance	1	2	5
Safety (x2) - Can be operated within a close distance, and is less likely to malfunction	3	3	3
Total	14	11	13

Design Matrix Justifications:

For mobility, the frisbee catapult was given a 5 because of its smaller size and the arm is detachable. The single flywheel launcher was scored a 3 and is less heavy compared to the double-flywheel launcher, scored a 2, which was much heavier and bigger due to addition of the conveyor belt and safety shield. For performance, the catapult was a 1 and doesn't launch the disc as far as the double flywheel launcher, which was scored a 5, because there is no motor-powered system. The single flywheel launcher was scored a 2 as the motor is smaller and has only one flywheel to launch the discs. For repairability, a score of 5 was given for the catapult since there are fewer parts and mechanisms that go into the design. Additionally, if anything were to break it would most likely be the tension mechanism which can easily be repaired and fixed. For the single-flywheel launcher, a 3 was given because it is a little more complicated with moving parts and a motor so repairing it would be more challenging than a catapult. Similarly, for the double-flywheel, a score of 3 was given as well because similar to the one-flywheel frisbee launcher, there are moving parts and a motor which might have to be replaced when damage occurs. For safety, a score of 3 was given for the catapult because sometimes if the bands -- which create the tension mechanism -- were to break, they can shoot out and injure someone. Likewise, a 3 was given for the single flywheel design due to the possibility of a frisbee launching out improperly. Finally, for the double-flywheel design, a 5 was given because it implemented a safety barrier to protect the frisbee if it were to malfunction.

Final CAD Design:

